A Tutorial of GPCCA Implementation to a Multi-modal Dataset

Preparation

Suppose the R package `GPCCA` is correctly installed. We attach it to the environment.

```
library(GPCCA)
```

Firstly, we briefly introduce the example multi-modal dataset provided by the package. `example_MultiModalData_list` is a synthetic multi-modal dataset with three modalities (a list that consists of 3 numeric matrices). There are six groups of samples within the dataset, with ground truth labels given by `example_MultiModalData_labels`. Artificial missing values are introduced, leading to an overall missing rate of 20%.

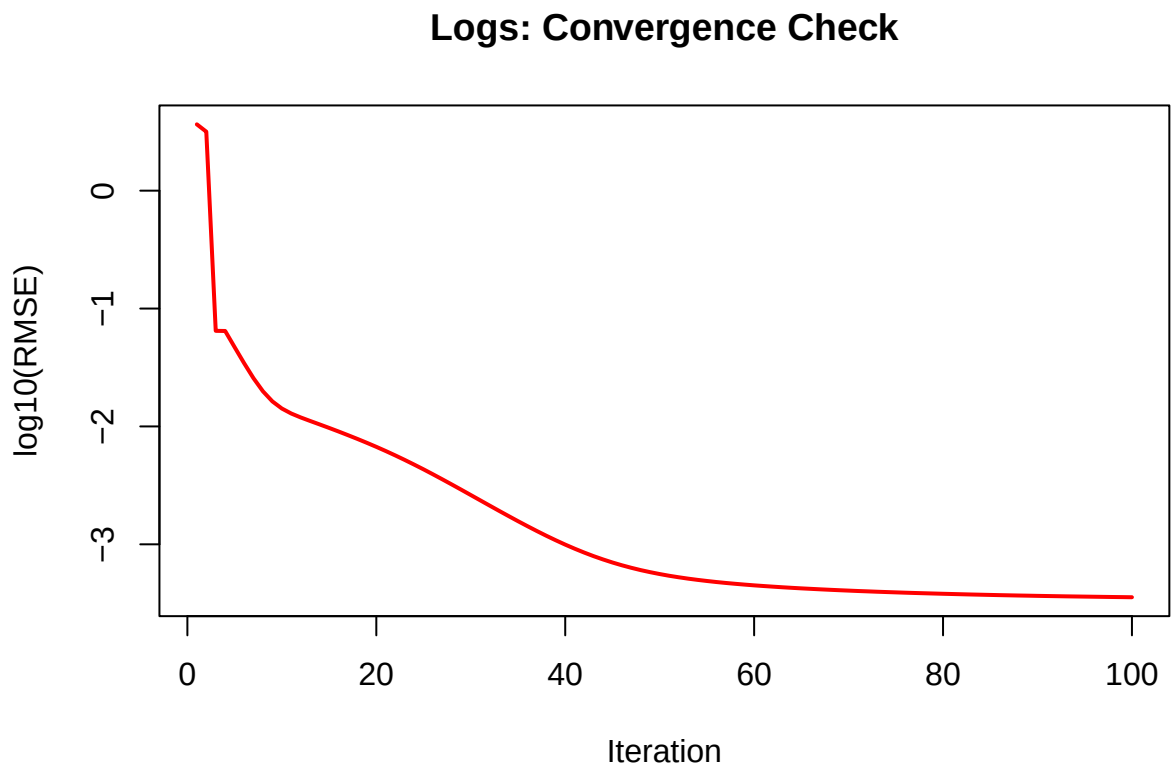
```
lbls <- example_MultiModalData_labels

str(example_MultiModalData_list)
```

```
## List of 3
##  $ M1: num [1:60, 1:600] 1.05 0.62 1.059 0.259 2.138 ...
##  $ M2: num [1:120, 1:600] NA NA 4.61 5.58 1.88 ...
##  $ M3: num [1:180, 1:600] -0.682 -1.904 -2.389 -3.201 -2.363 ...
```

Model Fit

Then, we fit GPCCA model to the example multi-modal dataset. We specify the size of target dimension to be `d = 5`, i.e. the number of latent factors in low-dimensional subspace. The default value of ridge regularization parameter `lambda = 0.5` is used. Here, we set `verbose = 0` to suppress the printing in every iteration.

```
GPCCA.fit <- GPCCAmodel(X.list = example_MultiModalData_list,
                        d = 5, lambda = 0.5, verbose = 0)
```



The backend of function `GPCCAmodel` is an EM algorithm. Once the model fitting is completed, we extract the fitted latent factors, i.e. the joint low-dimensional embeddings of the original multi-modal data.

```
LFs <- t(GPCCA.fit$Z)
```

Downstream Analysis

Next, we attach several more packages for our downstream clustering analysis and visualizations.

```
library(Seurat)
library(pdfCluster)
library(umap)
library(ggplot2)
```

To obtain the clustering result, we perform Louvain clustering on the joint low-dimensional embeddings learnt by GPCCA. The labels of identified clusters are stored in a factor object `grps`.

```
colnames(LFs) <- paste0("LF", 1:ncol(LFs))
rownames(LFs) <- paste0("sample", 1:nrow(LFs))

lc <- FindNeighbors(object = LFs, compute.SNN = TRUE, verbose = FALSE)
lc <- FindClusters(object = lc$snn, verbose = FALSE)

grps <- lc$res.0.8
```

To measure the agreement between the inferred cluster assignment and the true group labels. We compute the Adjusted Rand Index (ARI) and print the confusion matrix.

```r
adj.rand.index(lbls, grps)
```

```
## [1] 0.9382375
```

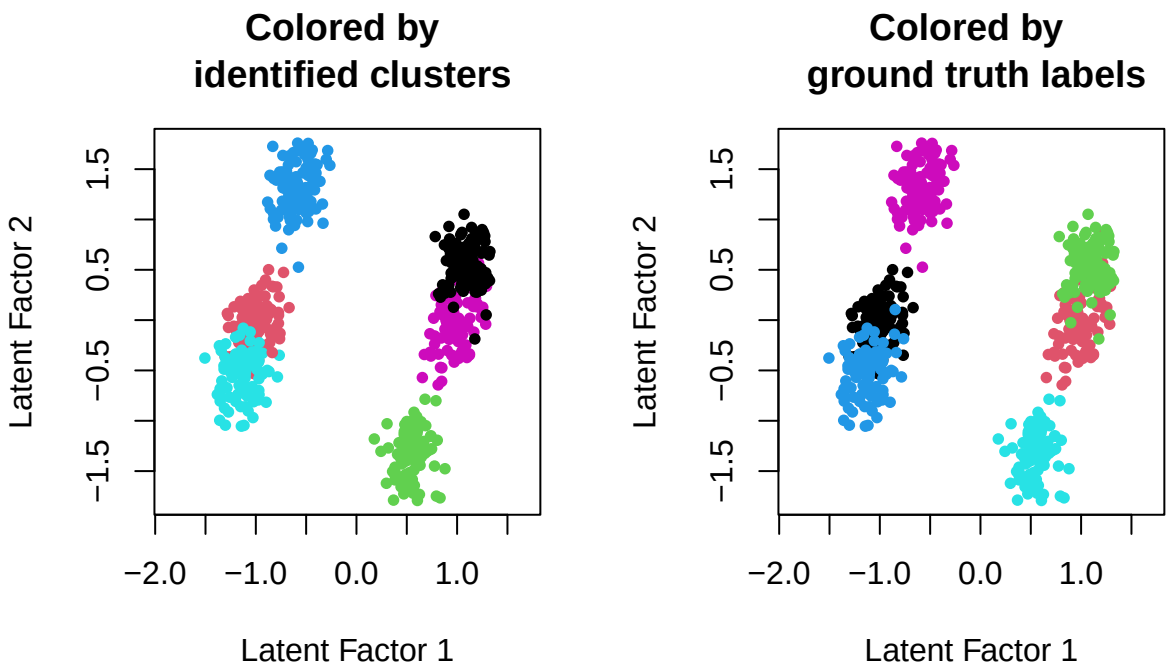
```r
table(lbls, grps)
```

```
##          grps
## lbls        0   1   2   3   4   5
##   group 1   0  98   0   0   2   0
##   group 2   8   0   0   0   0  92
##   group 3  98   0   0   0   0   2
##   group 4   0   4   0   0  96   0
##   group 5   0   0 100   0   0   0
##   group 6   0   0   0 100   0   0
```

Visualization

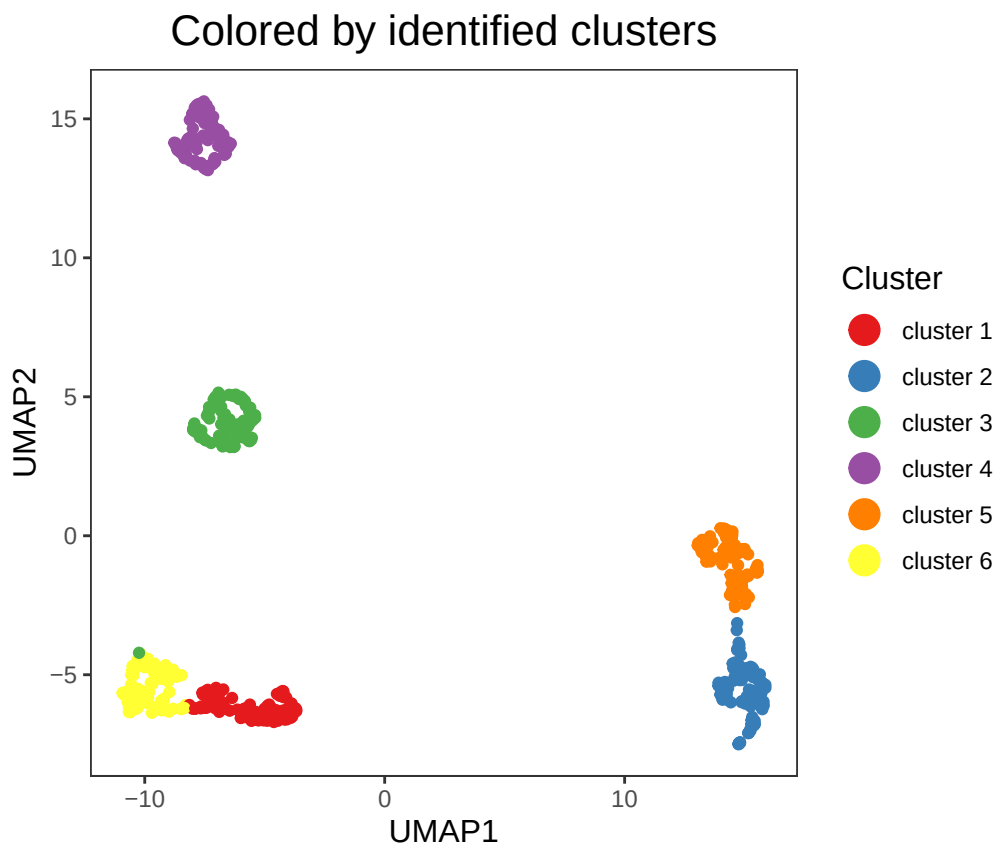
To visualize the fitted latent factors obtained by GPCCA, for example, we create a scatterplot of the first two latent factors.

```r
par(pty = "s")
par(mfrow = c(1, 2))
plot(x = LFs[, 1], y = LFs[, 2], col = grps,
     type = "p", pch = 16, cex = 0.8, asp = 1,
     xlab = "Latent Factor 1", ylab = "Latent Factor 2",
     main = "Colored by \n identified clusters")
plot(x = LFs[, 1], y = LFs[, 2], col = factor(lbls),
     type = "p", pch = 16, cex = 0.8, asp = 1,
     xlab = "Latent Factor 1", ylab = "Latent Factor 2",
     main = "Colored by \n ground truth labels")
```

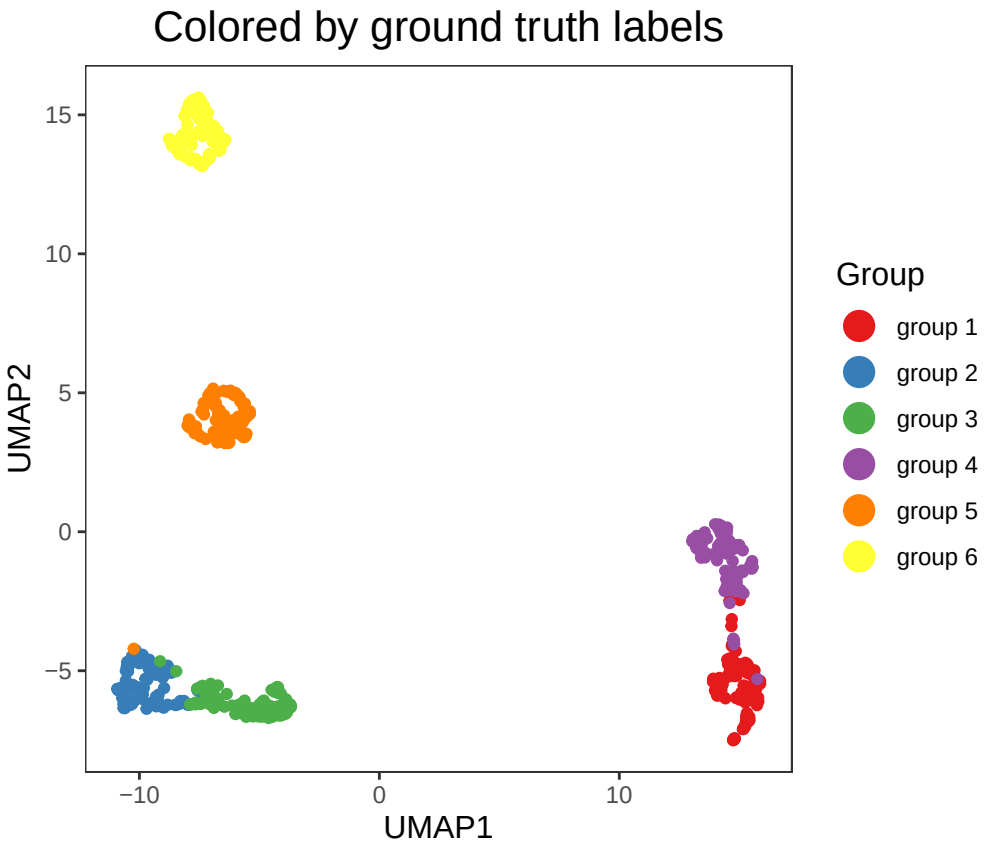


Furthermore, to visualize the potential of all joint low-dimensional embeddings combined, we create a UMAP based on the reconstructed data representations learnt by GPCCA.

```
umap_out <- umap(d = LFs)

plotDAT <- data.frame(UMAP1 = umap_out$layout[, 1], UMAP2 = umap_out$layout[, 2],
                      Cluster = as.factor(paste0("cluster ", as.numeric(grps))),
                      Group = as.factor(lbls))

ggplot(plotDAT, aes(x = UMAP1, y = UMAP2, color = Cluster)) + geom_point() + theme_bw() +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  scale_color_brewer(palette = "Set1") + theme(aspect.ratio = 1) +
  ggtitle("Colored by identified clusters") +
  theme(plot.title = element_text(hjust = 0.5, size = 16),
        axis.title = element_text(size = 12),
        legend.title = element_text(size = 12)) +
  guides(color = guide_legend(override.aes = list(size = 5)))
```




```
ggplot(plotDAT, aes(x = UMAP1, y = UMAP2, color = Group)) + geom_point() + theme_bw() +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank()) +
  scale_color_brewer(palette = "Set1") + theme(aspect.ratio = 1) +
  ggtitle("Colored by ground truth labels") +
  theme(plot.title = element_text(hjust = 0.5, size = 16),
        axis.title = element_text(size = 12),
        legend.title = element_text(size = 12)) +
  guides(color = guide_legend(override.aes = list(size = 5)))
```


Colored by ground truth labels
15
10
5
0
−5
UMAP2
−10
0
10
UMAP1
Group
group 1
group 2
group 3
group 4
group 5
group 6